

Topic 4

Acids and Bases

Chemistry in Context

Cutting onions One of the favourite party foods is *blooming onion*. Do you love it? It is easy to make a blooming onion at home. You can cut an onion into layers and create a shape that resembles a blooming flower with many petals. A blooming onion is obtained after you dip the onion in a batter and deep-fry it in oil. During the process, you probably notice that cutting raw onions makes you shed tears. Do you know why? The cells of an onion contain volatile sulphur compounds and other enzymes. When you cut through the cells of an onion, these compounds are released. They then mix to form sulfenic acids which in turn become gases. These gases irritate your eyes and make you shed tears.



Do you recall?

1 How can you distinguish between an acid and an alkali?
Making use of pH paper (or meter) / litmus paper (or solution) / universal indicator solution

2 What is the word equation for neutralization?
acid + alkali ⟶ salt + water

3 One mole of HCl(aq) is allowed to react with two moles of NaOH(aq). How many moles of water will be produced?
One mole

In this topic, you are going to study more about acids and alkalis.

blooming onion 洋葱花球

Unit

14 Acids and alkalis

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Characteristics of acids and alkalis
- Role of water for acids and alkalis

14.1 Acids in our daily lives

Many foods and drinks contain acids. For example, citrus fruits such as oranges, lemons and grapefruits contain **citric acid**. Soft drinks contain **carbonic acid**. All these substances are **acidic**.

Do you know that many household products also contain acids? The main ingredient in vinegar is **ethanoic acid**. Some toilet bowl cleansers contain **hydrochloric acid**. Our stomach wall secretes gastric juice with hydrochloric acid. Fig. 14.1 shows some items that contain acids.

Soft drinks may contain **phosphoric acid** and citric acid as well.

Hydrochloric acid in the stomach helps the breakdown of food, particularly protein.



(a) Citrus fruits contain citric acid



(b) Vinegar contains ethanoic acid



(c) Soft drinks contain carbonic acid N1



(d) This toilet bowl cleanser contains hydrochloric acid N2

Fig. 14.1 Some common items that contain acids

N1 Refer to page T15.

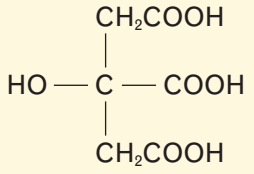
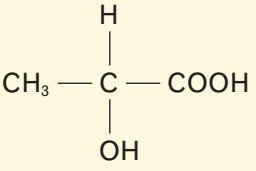
N2 Refer to page T15.

citric acid 檸檬酸　carbonic acid 碳酸　phosphoric acid 磷酸　acidic 酸性的　ethanoic acid 乙酸
hydrochloric acid 氫氯酸

Table 14.1 lists some common acids and their uses.

Table 14.1

Some common acids and their occurrences or uses

Acid	Chemical formula	Occurrence or use
Hydrochloric acid	CE 2006 Paper 2 Q11 HCl	• produced in the stomach • used to clean bricks, concrete and tiles N3
Sulphuric acid	CE 2009 Paper 2 Q14 H_2SO_4	• used in car batteries • used to make fertilizers CE 2006 Paper 2 Q11
Nitric acid	HNO_3	• used to make fertilizers
Ethanoic acid	CH_3COOH	• found in vinegar • used in *pickled* food (Fig. 14.2)
Citric acid		• found in citrus fruits • used as a flavour in drinks N4
Carbonic acid	H_2CO_3	• found in soft drinks
Lactic acid		• makes milk sour • builds up in muscles during *strenuous* exercise (Fig. 14.3)

N3 Metals tarnish because they react with the air to form a layer of oxide or sulphide. Acids can be used to remove this layer. One of the most difficult oxide layers to remove is rust. Rust removers usually contain an acid such as hydrochloric acid or phosphoric acid. The latter has the advantage that it also forms a protective phosphate layer to help prevent further corrosion.

Other metals can be cleaned using ethanoic or citric acid. An effective way of cleaning brass or copper is to rub it with a piece of lemon. Vinegar should not be used to clean copper and its alloys because of the formation of poisonous copper ethanoate. In all cases, always wash the metals well with water or washing soda after cleaning with acid. This is to remove possible toxic products and prevent the retarnishing which occurs if the metal is left in contact with acid.

N4 Benzoic acid is commonly found in canned grapefruit juice. Benzoic acid is toxic to bacteria. It is used as a preservative.



Fig. 14.2 These onions have been pickled in vinegar



Fig. 14.3 The painful *cramp* is caused by lactic acid built up in muscles during strenuous exercise

sulphuric acid 硫酸　nitric acid 硝酸　pickled 醃製　lactic acid 乳酸　strenuous 劇烈的　cramp 抽筋

14.2 Acids in the laboratory

Fig. 14.4 shows acids we often use in the laboratory:

- hydrochloric acid (HCl(aq));
- sulphuric acid (H_2SO_4(aq)); and
- nitric acid (HNO_3(aq)). N5

N5 Nitric acid is manufactured from ammonia.



Fig. 14.4 Common acids used in the laboratory: dilute hydrochloric acid, dilute sulphuric acid and dilute nitric acid

Hydrochloric acid is made by dissolving hydrogen chloride gas in water. Pure sulphuric acid and nitric acid are colourless liquids.

Concentrated acids contain acids dissolved in a small amount of water. Dilute acids contain acids dissolved in a large amount of water.

Diluting concentrated sulphuric acid

Caution: Wear safety glasses and protective gloves when diluting concentrated sulphuric acid. Carry out the process inside a fume cupboard.

Concentrated sulphuric acid reacts vigorously with water, giving out a great amount of heat. We should dilute the concentrated acid by adding it slowly to a large amount of water while stirring (Fig. 14.5).

CE 2007 Paper 1 Q9
ASL 2009 Paper 1 Q4(a)
AL 2009 Paper 1 Q7(c)

N6 Examination questions often ask students to explain why water should NOT be added to concentrated sulphuric acid during dilution.

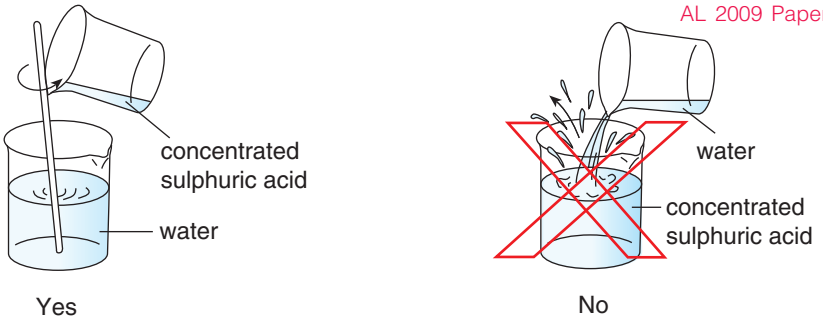


Fig. 14.5 Dilute concentrated sulphuric acid by adding it slowly to a large amount of water while stirring N6

We should NEVER add water to the concentrated acid. When you add water to the acid, so much heat is released that the solution may violently *splash* out of the container and *sputter* onto your face.

Video
Investigating the properties of dilute acid

14.1
Investigating the properties of dilute hydrochloric acid.

14.3 Characteristics of dilute acids

Dilute acids, such as dilute hydrochloric acid and dilute sulphuric acid, show a range of common properties.

Taste

Most dilute acids have a sour taste.

NEVER taste any acid in the laboratory!

Effect on indicators

We can use an **indicator** to identify whether a colourless solution is acidic or alkaline. An indicator is a substance (usually a vegetable dye) which gives different colours in acidic and alkaline solutions.

Most acids are colourless.

We will discuss alkaline solutions later in this unit.

Litmus solution, **methyl orange** and **phenolphthalein** are common indicators used in the laboratory (Fig. 14.6). Fig. 14.7 and Table 14.2 show the colours of these indicators in an acid.

Litmus paper can be made by soaking a piece of filter paper in litmus solution and then drying it. There are two types of litmus paper: red litmus paper and blue litmus paper. Red litmus paper turns blue in alkaline solutions. Blue litmus paper turns red in acidic solutions.



Fig. 14.6 Some common indicators: litmus solution, methyl orange and phenolphthalein



Fig. 14.7 Colours of litmus solution, methyl orange and phenolphthalein in hydrochloric acid (from left to right)

splash 濺出　sputter 噴濺　indicator 指示劑　litmus solution 石蕊試液　litmus paper 石蕊試紙
methyl orange 甲基橙試液　phenolphthalein 酚酞試液

Table 14.2

Colours of indicators in an acid	
Indicator	**Colour in an acid**
Litmus solution	red
Methyl orange	red
Phenolphthalein	colourless

Do you know ?

What is the colour of hydrangea flowers in acidic and alkaline soils?

Hydrangea flowers contain natural indicators — they are blue if grown on acidic soil and pink if grown on alkaline soil.



Fig. 14.8 Different colours of hydrangea flowers in acidic (left) and alkaline (right) soils

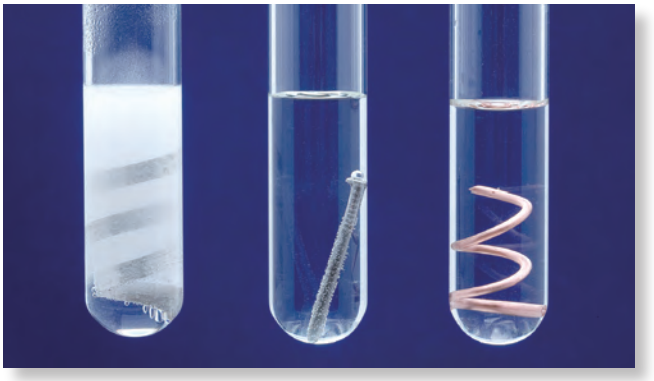
Reaction with metals

Dilute hydrochloric acid, dilute sulphuric acid and very dilute nitric acid◂ react with reactive metals◂ (such as magnesium, zinc and iron) to give hydrogen gas. Unreactive metals (such as copper and silver) do not react with the acids (Fig. 14.9).

DSE 2012 Paper 1A Q19 CE 2005 Paper 2 Q38

◂ Only very dilute nitric acid reacts with reactive metals to give hydrogen gas. Fairly dilute nitric acid, however, produces nitrogen monoxide gas (NO) when it reacts with metals. We will discuss this reaction in Topic 5 Redox Reactions, Chemical Cells and Electrolysis.

◂ Potassium and sodium react explosively with dilute acids. NEVER do these reactions in the laboratory.



(a) Magnesium reacts rapidly (b) Iron reacts slowly (c) Copper does not react

Fig. 14.9 Reactions of dilute hydrochloric acid with magnesium, iron and copper

hydrangea flower 繡球花

✓ metal + dilute acid ⟶ salt + hydrogen

CE 2008 Paper 1 Q4(a)

For example, magnesium reacts with dilute hydrochloric acid, dilute sulphuric acid or very dilute nitric acid to give hydrogen gas. This indicates that hydrogen ions are present in dilute acids.

$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$ CE 2008 Paper 1 Q4(a)

$Mg(s) + H_2SO_4(aq) \longrightarrow MgSO_4(aq) + H_2(g)$

$Mg(s) + 2HNO_3(aq) \longrightarrow Mg(NO_3)_2(aq) + H_2(g)$

We can represent these reactions by the following ionic equation:

$Mg(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2(g)$

Acids at home, such as vinegar and lemon juice, react with magnesium to give hydrogen gas too.

Reaction with carbonates

CE 2005 Paper 2 Q38
CE 2006 Paper 2 Q48

Dilute acids react with carbonates to give carbon dioxide gas.

✓ carbonate + dilute acid ⟶ salt + water + carbon dioxide

Sodium carbonate (solid or aqueous solution) reacts with dilute acids to give carbon dioxide gas. *Effervescence* occurs as a result (Fig. 14.10).



Fig. 14.10 Sodium carbonate reacts with dilute hydrochloric acid to give carbon dioxide gas. Effervescence occurs as a result

effervescence 泡騰

$Na_2CO_3(s \text{ or } aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$

$Na_2CO_3(s \text{ or } aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + H_2O(l) + CO_2(g)$

$Na_2CO_3(s \text{ or } aq) + 2HNO_3(aq) \longrightarrow 2NaNO_3(aq) + H_2O(l) + CO_2(g)$

We can represent the reactions by the following ionic equation:

$CO_3^{2-}(aq) + 2H^+(aq) \longrightarrow H_2O(l) + CO_2(g)$

Solid Na_2CO_3 dissolves in water to give $CO_3^{2-}(aq)$ ions.

Calcium chloride and calcium nitrate are soluble in water. However, when dilute sulphuric acid reacts with calcium carbonate, insoluble calcium sulphate forms. The calcium sulphate covers the surface of calcium carbonate and prevents further reaction. N7

Do not use $CO_3^{2-}(aq)$ to represent $CaCO_3(s)$ because calcium carbonate is insoluble in water and there is no separate CO_3^{2-} ion in water.

CE 2006 Paper 2 Q48
DSE 2013 Paper 1A Q19

N7 Reacting calcium carbonate with dilute sulphuric acid is NOT an appropriate method for preparing carbon dioxide.

Calcium carbonate reacts with dilute hydrochloric acid and dilute nitric acid to form calcium salts. Carbon dioxide gas is formed in the reactions.

CE 2009 Paper 1 Q1(b)

$CaCO_3(s) + 2HCl(aq) \longrightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

$CaCO_3(s) + 2HNO_3(aq) \longrightarrow Ca(NO_3)_2(aq) + H_2O(l) + CO_2(g)$

We can represent the reactions by the following ionic equation:

$CaCO_3(s) + 2H^+(aq) \longrightarrow Ca^{2+}(aq) + H_2O(l) + CO_2(g)$

CE 2009 Paper 1 Q1(b)

We can use limewater to test for carbon dioxide gas. Carbon dioxide turns limewater milky.

Reaction with hydrogencarbonates

Dilute acids react with hydrogencarbonates to give carbon dioxide gas.

✓ hydrogencarbonate + dilute acid $\longrightarrow$ salt + water + carbon dioxide

Sodium hydrogencarbonate (solid or aqueous solution) reacts with acids to give carbon dioxide gas. Effervescence also occurs.

$NaHCO_3(s \text{ or } aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$

$2NaHCO_3(s \text{ or } aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + 2H_2O(l) + 2CO_2(g)$

$NaHCO_3(s \text{ or } aq) + HNO_3(aq) \longrightarrow NaNO_3(aq) + H_2O(l) + CO_2(g)$

We can represent the reactions by the following ionic equation:

$HCO_3^-(aq) + H^+(aq) \longrightarrow H_2O(l) + CO_2(g)$

Reaction with hydroxides and oxides of metals

We will further discuss neutralization in Unit 16.

Hydroxides and oxides of metals are called **bases**. When an acid reacts with a base, a salt and water are formed. Both the acid and the base lose their typical properties after the reaction. The reaction is called **neutralization**. N8

CE 2005 Paper 2 Q38

N8 Refer to page T16.

N9 Both oxides and hydroxides do NOT give a gas with acids.

✓ acid + metal hydroxide ⟶ salt + water

✓ acid + metal oxide ⟶ salt + water N9

Dilute hydrochloric acid reacts with magnesium hydroxide to give magnesium chloride and water.

$$2HCl(aq) + Mg(OH)_2(s) \longrightarrow MgCl_2(aq) + 2H_2O(l)$$

Dilute sulphuric acid reacts with copper(II) oxide to give copper(II) sulphate and water (Fig. 14.11).

$$H_2SO_4(aq) + CuO(s) \longrightarrow CuSO_4(aq) + H_2O(l)$$



Adding dilute sulphuric acid to copper(II) oxide



Heating the reaction mixture to give a blue copper(II) sulphate solution

Fig. 14.11 Copper(II) oxide reacts with dilute sulphuric acid to give a blue copper(II) sulphate solution

base 鹽基　　neutralization 中和作用

Electrical conductivity

All dilute acids conduct electricity due to the presence of mobile ions. We can use the experimental set-up shown in Fig. 14.12 to test the electrical conductivity of a dilute acid.

CE 2011 Paper 2 Q29

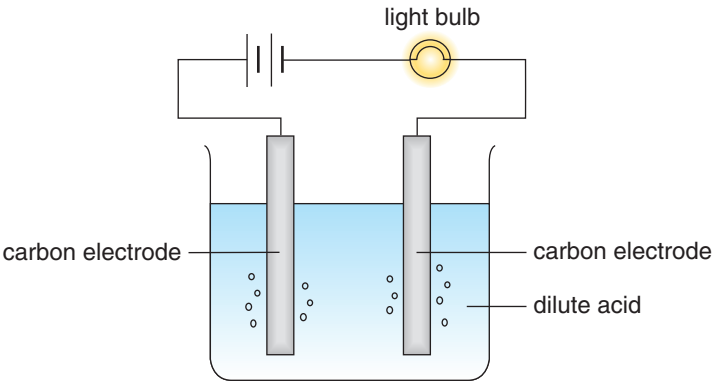


Fig. 14.12 Experimental set-up to test the electrical conductivity of a dilute acid

Practice 14.1

Consider the following experiments:

1 Solid zinc oxide is added to dilute nitric acid.

2 Ammonium carbonate solution is added to dilute hydrochloric acid.

For each of the experiments:

a) state the observable changes;

b) write a chemical equation; and

c) write an ionic equation for the reaction involved.

1 a) The oxide dissolves in the acid.
b) $ZnO(s) + 2HNO_3(aq) \longrightarrow Zn(NO_3)_2(aq) + H_2O(l)$
c) $ZnO(s) + 2H^+(aq) \longrightarrow Zn^{2+}(aq) + H_2O(l)$
2 a) Effervescence occurs.
b) $(NH_4)_2CO_3(aq) + 2HCl(aq) \longrightarrow 2NH_4Cl(aq) + H_2O(l) + CO_2(g)$
c) $CO_3^{2-}(aq) + 2H^+(aq) \longrightarrow H_2O(l) + CO_2(g)$

14.4 The role of water for acids

14.2
Comparing the properties of solid citric acid and its aqueous solution.

Video
Studying the role of water in exhibiting the acidic properties of citric acid

So far we have only considered characteristics of acids in aqueous solutions. What will happen if water is absent? Let's use citric acid as an example (Fig. 14.13).

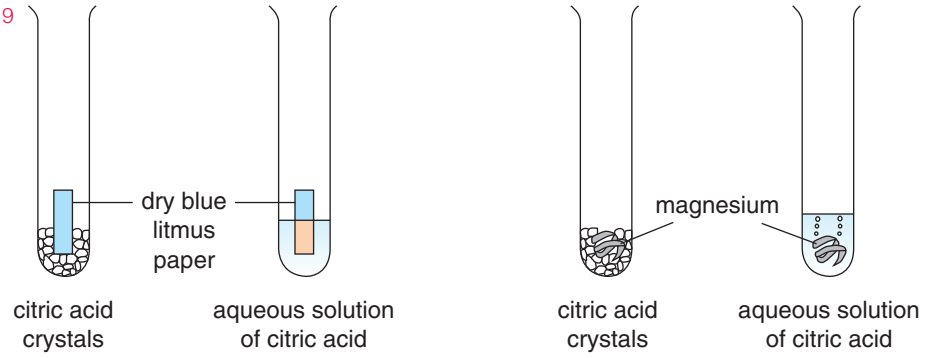


Fig. 14.13 Citric acid crystals

N10 Refer to page T17.

Citric acid crystals have no effect on dry blue litmus paper and magnesium. However, when we dissolve the crystals in water, the solution turns dry blue litmus paper red and reacts with magnesium to give hydrogen gas (Fig. 14.14). N10

CE 2006 Paper 2 Q28
CE 2010 Paper 2 Q28
DSE 2012 Paper 1A Q19



(a) Testing with dry blue litmus paper (b) Testing with magnesium

Fig. 14.14 Testing citric acid crystals and an aqueous solution of citric acid

CE 2005 Paper 2 Q29

The aqueous solution of citric acid shows typical properties of an acid while the citric acid crystals do not. This suggests that water must be present for an acid to show its acidic properties.

When citric acid crystals dissolve in water, the molecules dissociate (or ionize) to give hydrogen ions ($H^+(aq)$). This process is called **dissociation**.

Ionization refers to the formation of ions from atoms or molecules.

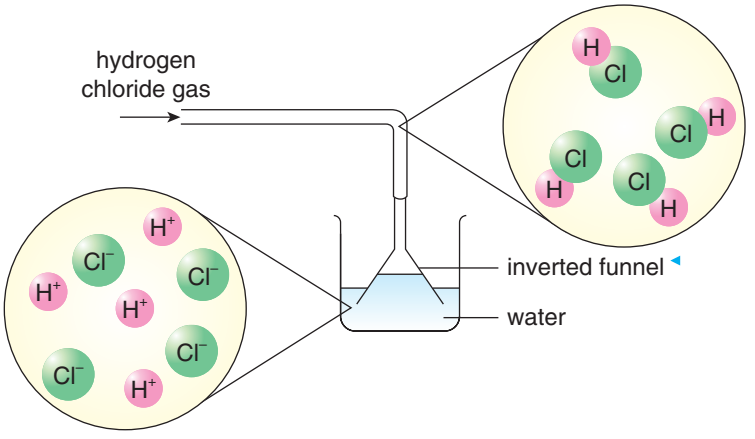
$$\text{citric acid(s)} + \text{water} \longrightarrow H^+(aq) + \text{citrate ion(aq)}$$

Hydrogen ions are responsible for all the acidic properties.

CE 2006 Paper 2 Q28

When hydrogen chloride gas is dissolved in water, hydrogen chloride molecules dissociate to give hydrogen ions (Fig. 14.15).

$$HCl(g) + \text{water} \longrightarrow H^+(aq) + Cl^-(aq)$$



An inverted funnel is used to prevent sucking back because hydrogen chloride gas is very soluble in water.

DSE 2012 Paper 1B Q7(c)

Fig. 14.15 The HCl molecules dissociate in water to form ions (This is only a schematic sketch. It does not show the actual numbers and sizes of particles in the acid; the ions are colourless.)

dissociation 離解作用

Do you know ?

Baking powder

Baking powder is a mixture of sodium hydrogencarbonate (baking soda) and a solid acid.

When we make cakes, we first add water and a little baking powder to a measured amount of flour. The cake mixture is then heated in an oven.

When sodium hydrogencarbonate is heated, it decomposes to give carbon dioxide gas:

$2NaHCO_3(s) \longrightarrow Na_2CO_3(s) + CO_2(g) + H_2O(l)$

The solid acid in the powder gives hydrogen ions when dissolved in water. Carbon dioxide gas is also liberated by the action of hydrogen ions on sodium hydrogencarbonate. The carbon dioxide gas makes the cake 'rise'.

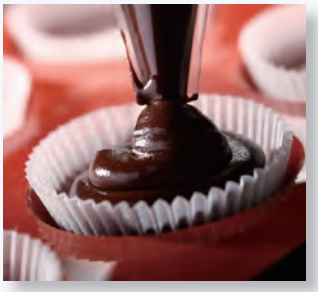


Fig. 14.16 Making cakes

Pure sulphuric acid and nitric acid are colourless liquids consisting of molecules. They do not show acidic properties because no hydrogen ions are present. When the two acids are dissolved in water, hydrogen ions are formed. They show the typical properties of an acid.

$H_2SO_4(l) + water \longrightarrow 2H^+(aq) + SO_4^{2-}(aq)$

$HNO_3(l) + water \longrightarrow H^+(aq) + NO_3^-(aq)$

We can give the definition of an acid in the following way:

✓ An acid is a hydrogen-containing substance that gives hydrogen ions ($H^+(aq)$) as the only type of positive ions when dissolved in water.

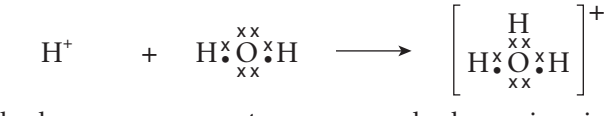
We will see that the dissociation of sulphuric acid occurs in two steps later in this unit. Equations representing the two steps are combined here for simplicity.

Hydrogen ion in aqueous solution

A hydrogen ion is simply a proton and it is too reactive to exist independently in an aqueous solution. In an aqueous solution, a hydrogen ion forms a dative covalent bond with a water molecule to give a **hydroxonium ion** (or **hydronium ion**), H_3O^+.

We have discussed dative covalent bonds in Topic 2 Microscopic World I.

Animation
Dissolving hydrogen chloride in water



hydrogen ion | water molecule | hydroxonium ion (or hydronium ion)

When writing equations, we usually represent a hydrogen ion in an aqueous solution as $H^+(aq)$ for simplicity.

hydroxonium ion (or hydronium ion) 水合氫離子

Practice 14.2

Look at the ingredients of Alka-Seltzer.



Each tablet contains: aspirin 324 mg as active ingredient. Each tablet also contains sodium hydrogencarbonate 1.737 g, citric acid 1.233 g.

When the tablet is added to water, the citric acid in the tablet dissociates in water to give hydrogen ions.
The sodium hydrogencarbonate in the tablet dissolves in water, releasing hydrogencarbonate ions.
The hydrogen ions react with the hydrogencarbonate ions, forming carbon dioxide gas.
$H^+(aq) + HCO_3^-(aq) \longrightarrow H_2O(l) + CO_2(g)$

Explain why effervescence occurs when we drop a tablet of Alka-Seltzer in water. Write an ionic equation for the reaction involved.

14.5 Basicity of an acid

✓ The maximum number of hydrogen ions produced by an acid molecule is called the **basicity** of the acid.

Every hydrogen chloride molecule dissociates in water to give one hydrogen ion and one chloride ion. Thus, the basicity of hydrochloric acid is 1. It is a **monobasic acid**.

CE 2005 Paper 2 Q14

$$HCl(aq) \longrightarrow H^+(aq) + Cl^-(aq)$$

Every sulphuric acid molecule dissociates in water to give two hydrogen ions. Thus, the basicity of sulphuric acid is 2. It is a **dibasic acid**. The dissociation of sulphuric acid occurs in two steps:

CE 2005 Paper 2 Q14

$$H_2SO_4(aq) \longrightarrow H^+(aq) + HSO_4^-(aq)$$

$$HSO_4^-(aq) \rightleftharpoons H^+(aq) + SO_4^{2-}(aq)$$

A single arrow ($\longrightarrow$) indicates complete dissociation of the acid. A double arrow ($\rightleftharpoons$) indicates partial dissociation of the acid.

basicity 鹽基度　　monobasic acid 一元酸　　dibasic acid 二元酸

Phosphoric acid is a **tribasic acid**. Equations for the successive dissociations of phosphoric acid are as follows:

$$H_3PO_4(aq) \rightleftharpoons H^+(aq) + H_2PO_4^-(aq)$$

$$H_2PO_4^-(aq) \rightleftharpoons H^+(aq) + HPO_4^{2-}(aq)$$

$$HPO_4^{2-}(aq) \rightleftharpoons H^+(aq) + PO_4^{3-}(aq)$$

We can also say that only the hydrogen atom in the –COOH group is 'ionizable'.

It is often difficult to determine the basicity of an acid just from its chemical formula. For example, a molecule of ethanoic acid (CH_3COOH) contains four hydrogen atoms but only the hydrogen atom in the –COOH group can undergo dissociation. Thus, it is a monobasic acid (Fig. 14.17).

CE 2005 Paper 2 Q14

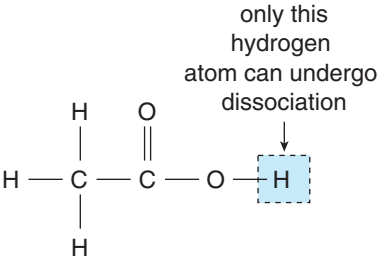


Fig. 14.17 Ethanoic acid is a monobasic acid because only the hydrogen atom in the –COOH group can undergo dissociation

14.6 Bases and alkalis

Bases are substances which neutralize acids. Bases are usually oxides or hydroxides of metals. When a base neutralizes an acid, only a salt and water are formed.

✓ A base is a compound which reacts with an acid to give a salt and water only.

✓ acid + base ⟶ salt + water

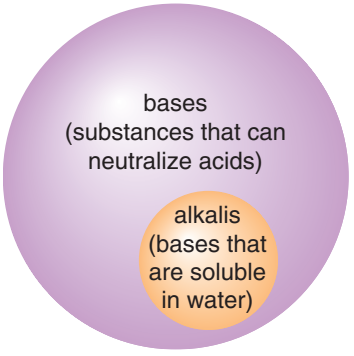


Fig. 14.18 Bases and alkalis

An **alkali** is a special kind of base — one that is soluble in water (Fig. 14.18). Sodium hydroxide, potassium hydroxide, calcium hydroxide and sodium oxide are alkalis. Ammonia gas dissolves in water to form an **alkaline** solution. Thus, aqueous ammonia is also regarded as an alkali.

tribasic acid 三元酸　　alkali 鹼　　alkaline 鹼性的

We will discuss soap in Topic 8 Chemistry of Carbon Compounds.



Fig. 14.19 A drain cleanser contains sodium hydroxide

N11 Refer to page T18.

14.7 Bases and alkalis in the home

Alkalis such as sodium hydroxide and aqueous ammonia are good household cleansers because alkalis react with fats to form soap (Fig. 14.19). Bases such as aluminium hydroxide and magnesium hydroxide are used as active ingredients in *antacids* for neutralizing the excess acid in the stomach.

Table 14.3 lists some common bases and alkalis and their uses.

Table 14.3

Some common bases and alkalis and their uses

Base / Alkali	Chemical formula	Use
Sodium hydroxide (*caustic soda*)	NaOH	as household oven cleanser and drain cleanser N11
Ammonia	NH_3	as household glass cleanser
Calcium hydroxide (slaked lime)	$Ca(OH)_2$	to neutralize **acidity** in soil
Aluminium hydroxide and magnesium hydroxide	$Al(OH)_3$ $Mg(OH)_2$	as active ingredients in antacids

14.8 Alkalis in the laboratory

Fig. 14.20 shows solutions of four alkalis we often use in the laboratory:

- potassium hydroxide solution, KOH(aq);
- sodium hydroxide solution, NaOH(aq);
- calcium hydroxide solution, $Ca(OH)_2$(aq); and
- aqueous ammonia, NH_3(aq).



Fig. 14.20 Common alkalis used in the laboratory: dilute potassium hydroxide solution, dilute sodium hydroxide solution, calcium hydroxide solution and aqueous ammonia

antacid 制酸劑 caustic soda 苛性鈉 acidity 酸度

Video
Investigating the properties of dilute solutions of alkalis

14.3
Investigating the properties of dilute solutions of alkalis.

NEVER taste any alkali in the laboratory!

Reference Website
Refer to the video clip 'Acids and Alkalis': http://resources.hkedcity.net/resource_detail.php?rid=1851701568 (accessed July 2014)

14.9 Characteristics of solutions of alkalis

Solutions of alkalis show a range of common properties.

Taste

Solutions of alkalis have a bitter taste.

Feel

Dilute solutions of alkalis have a 'slippery' feel.

Effect on indicators

Fig. 14.21 and Table 14.4 show the colours of litmus solution, methyl orange and phenolphthalein in the solution of an alkali.



Fig. 14.21 Colours of litmus solution, methyl orange and phenolphthalein in sodium hydroxide solution (from left to right)

Table 14.4

Colours of indicators in the solution of an alkali	
Indicator	**Colour in the solution of an alkali**
Litmus solution	blue — DSE 2014 Paper 1A Q21
Methyl orange	yellow
Phenolphthalein	red

Reaction with metal ions

Reaction of dilute sodium hydroxide solution with metal ions

Most metal hydroxides are insoluble. Table 14.5 lists some common metal hydroxide precipitates formed when we add dilute sodium hydroxide solution to solutions containing the metal ions.

The specific colours of the precipitates may help us identify the metal ions in the solutions. Such identification is an important test in analytical chemistry.

N12 Refer to page T19.

Table 14.5

Some common metal hydroxide precipitates formed upon the addition of dilute sodium hydroxide solution and the ionic equations involved N12

Adding NaOH(aq) to solution containing	Metal hydroxide precipitate formed	Colour of the precipitate	Ionic equation involved
$Ca^{2+}(aq)$	$Ca(OH)_2(s)$	white	$Ca^{2+}(aq) + 2OH^-(aq) \longrightarrow Ca(OH)_2(s)$
$Mg^{2+}(aq)$	$Mg(OH)_2(s)$	white	$Mg^{2+}(aq) + 2OH^-(aq) \longrightarrow Mg(OH)_2(s)$
$Al^{3+}(aq)$	$Al(OH)_3(s)$	white	$Al^{3+}(aq) + 3OH^-(aq) \longrightarrow Al(OH)_3(s)$
$Pb^{2+}(aq)$	$Pb(OH)_2(s)$	white	$Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$
$Zn^{2+}(aq)$	$Zn(OH)_2(s)$	white	$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$
$Fe^{2+}(aq)$	$Fe(OH)_2(s)$	green	$Fe^{2+}(aq) + 2OH^-(aq) \longrightarrow Fe(OH)_2(s)$
$Fe^{3+}(aq)$	$Fe(OH)_3(s)$	reddish brown	$Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$
$Cu^{2+}(aq)$	$Cu(OH)_2(s)$	pale blue	$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$

CE 2010 Paper 2 Q20
DSE 2013 Paper 1A Q3
ASL 2011 Paper 2 Q4(b)
AL 2011 Paper 2 Q6(c)
CE 2009 Paper 2 Q37
N13
DSE 2013 Paper 1A Q9
DSE 2014 Paper 1B Q9(a)

Calcium hydroxide is slightly soluble in water.

Fig. 14.22 shows the appearance of some metal hydroxide precipitates.



calcium hydroxide | magnesium hydroxide | aluminium hydroxide | lead(II) hydroxide | zinc hydroxide | iron(II) hydroxide | iron(III) hydroxide | copper(II) hydroxide

Fig. 14.22 Appearance of some common metal hydroxide precipitates

N13 Examination questions often ask about the action of NaOH(aq) / KOH(aq) on $Fe^{2+}(aq)$ ions and $Fe^{3+}(aq)$ ions. Students need to memorize the colours of the precipitates formed.

Aluminium hydroxide, lead(II) hydroxide and zinc hydroxide react with excess dilute sodium hydroxide solution to form soluble **complex salts**. Colourless solutions are formed as a result (Fig. 14.23).

AL 2011 Paper 1 Q7(b)(i)

$$Al(OH)_3(s) + OH^-(aq) \longrightarrow [Al(OH)_4]^-(aq)$$

tetrahydroxoaluminate ion (colourless solution)

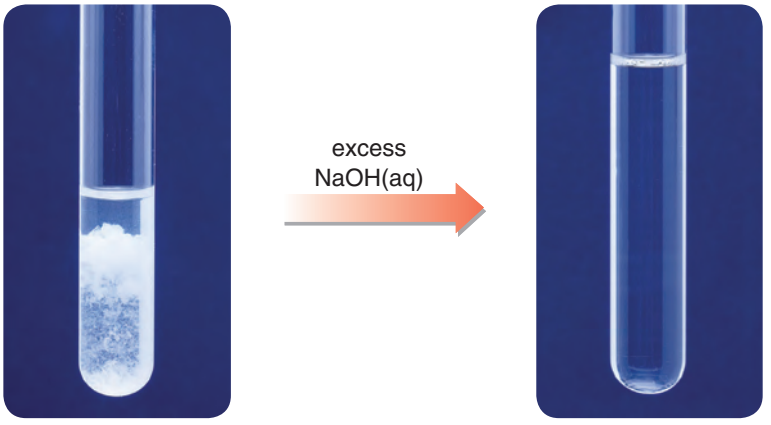
ASL 2011 Paper 2 Q4(b)
AL 2011 Paper 2 Q6(c)

$$Pb(OH)_2(s) + 2OH^-(aq) \longrightarrow [Pb(OH)_4]^{2-}(aq)$$

tetrahydroxoplumbate(II) ion (colourless solution)

$$Zn(OH)_2(s) + 2OH^-(aq) \longrightarrow [Zn(OH)_4]^{2-}(aq)$$

tetrahydroxozincate ion (colourless solution)



(a) Aluminium hydroxide precipitate

(b) A colourless solution formed

Fig. 14.23 Aluminium hydroxide reacts with excess dilute sodium hydroxide solution to form a colourless solution

Reaction of dilute aqueous ammonia with metal ions

When ammonia gas dissolves in water, it reacts with water to give ammonium ions ($NH_4^+(aq)$) and hydroxide ions ($OH^-(aq)$).

$$NH_3(g) + \text{water} \longrightarrow NH_3(aq)$$

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

complex salt 絡鹽

Solutions containing metal ions also form hydroxide precipitates with dilute aqueous ammonia. Some common examples are listed in Table 14.6.

Table 14.6

Some common metal hydroxide precipitates formed upon the addition of dilute aqueous ammonia and the ionic equations involved

Adding $NH_3(aq)$ to solution containing	Metal hydroxide precipitate formed	Colour of the precipitate	Ionic equation involved
$Mg^{2+}(aq)$	$Mg(OH)_2(s)$	white	$Mg^{2+}(aq) + 2OH^-(aq) \longrightarrow Mg(OH)_2(s)$
$Al^{3+}(aq)$	$Al(OH)_3(s)$	white	$Al^{3+}(aq) + 3OH^-(aq) \longrightarrow Al(OH)_3(s)$
$Pb^{2+}(aq)$	$Pb(OH)_2(s)$	white	$Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$
$Zn^{2+}(aq)$	$Zn(OH)_2(s)$	white	$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$
$Fe^{2+}(aq)$	$Fe(OH)_2(s)$	green	$Fe^{2+}(aq) + 2OH^-(aq) \longrightarrow Fe(OH)_2(s)$
$Fe^{3+}(aq)$	$Fe(OH)_3(s)$	reddish brown	$Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$
$Cu^{2+}(aq)$	$Cu(OH)_2(s)$	pale blue	$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$

DSE 2013 Paper 1A Q11 DSE 2012 Paper 1A Q20 AL 2010 Paper 1 Q3(b)(ii)

A solution containing $Ca^{2+}(aq)$ ions does not give a precipitate when mixed with dilute aqueous ammonia as the concentration of $OH^-(aq)$ ions in dilute aqueous ammonia is low.

The $[Zn(NH_3)_4]^{2+}(aq)$ ion has the following name since there are 4 ammonia molecules surrounding the zinc ion:

tetraamminezinc ion

- tetra: indicating 4 ammonia molecules are present
- ammine: indicating ammonia molecule is present

Zinc hydroxide and copper(II) hydroxide react with excess dilute aqueous ammonia to form soluble complex salts. Zinc hydroxide gives a colourless solution (Fig. 14.24) while copper(II) hydroxide gives a deep blue solution (Fig. 14.25).

$$Zn(OH)_2(s) + 4NH_3(aq) \longrightarrow [Zn(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$$

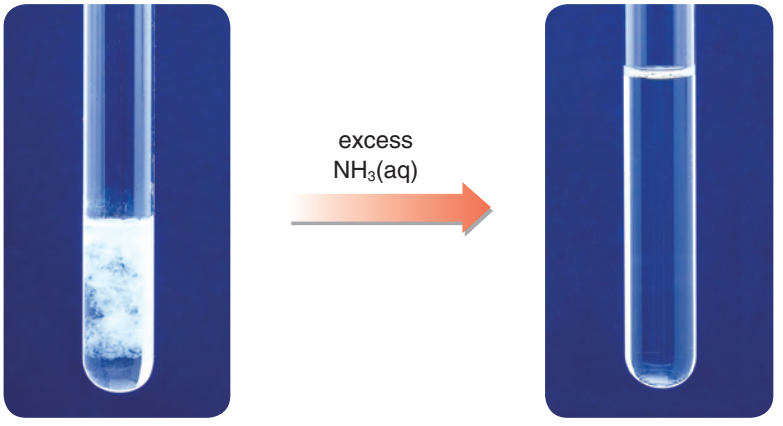
tetraamminezinc ion (colourless solution)

DSE 2014 Paper 1A Q21 AL 2010 Paper 1 Q3(b)(ii)

AL 2009 Paper 1 Q3(a)(iii) AL 2010 Paper 1 Q3(b)(ii)

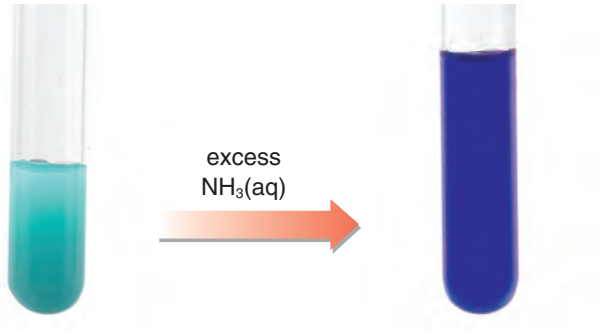
$$Cu(OH)_2(s) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$$

tetraamminecopper(II) ion (deep blue solution)



(a) Zinc hydroxide precipitate (b) A colourless solution formed

Fig. 14.24 Zinc hydroxide reacts with excess dilute aqueous ammonia to form a colourless solution



(a) Copper(II) hydroxide precipitate

(b) A deep blue solution formed

Fig. 14.25 Copper(II) hydroxide reacts with excess dilute aqueous ammonia to form a deep blue solution

Video
Action of solid calcium hydroxide on ammonium chloride

Reaction with ammonium compounds

Ammonium chloride (NH_4Cl), ammonium nitrate (NH_4NO_3) and ammonium sulphate ($(NH_4)_2SO_4$) all contain ammonium ions. N14

Heating solids or solutions of ammonium compounds with solutions of alkalis liberates ammonia gas. N15

CE 2010 Paper 1 Q2(a)(ii)
CE 2010 Paper 2 Q20
DSE 2013 Paper 1A Q9

$$2NH_4Cl(s \text{ or } aq) + Ca(OH)_2(aq) \xrightarrow{\text{heat}} 2NH_3(g) + CaCl_2(aq) + 2H_2O(l)$$

$$NH_4Cl(s \text{ or } aq) + NaOH(aq) \xrightarrow{\text{heat}} NH_3(g) + NaCl(aq) + H_2O(l)$$

We can also represent the reactions by the following ionic equation:

$$NH_4^+(aq) + OH^-(aq) \xrightarrow{\text{heat}} NH_3(g) + H_2O(l)$$

DSE 2012 Paper 1B Q7(a)

Reaction with acid

A solution of an alkali reacts with an acid to give a salt and water only.

For example, dilute sodium hydroxide solution reacts with dilute hydrochloric acid to give sodium chloride and water.

$$NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$$

Ammonia gas has a *pungent* smell. We can test the ammonia gas with moist red litmus paper. It turns the paper blue.

N14 Be careful with the spelling of the chemicals 'ammonium chloride', 'ammonium sulphate', etc.

N15 Examination questions often ask about the chemical test for identifying an ammonium compound. Remember it is necessary to heat the sample under test with sodium hydroxide solution / calcium hydroxide solution. The ammonium compound gives an alkaline gas, NH_3.

pungent 辛辣的

Electrical conductivity

All solutions of alkalis conduct electricity due to the presence of mobile ions.

CE 2011 Paper 2 Q12

14.10 The role of water for alkalis

Potassium hydroxide and sodium hydroxide solids both consist of ions. These ions (K^+, Na^+ and OH^-) are not mobile. When these solids dissolve in water, they give mobile hydroxide ions ($OH^-(aq)$) which are the only type of negative ions in the solution.

$$KOH(s) + water \longrightarrow K^+(aq) + OH^-(aq)$$

$$NaOH(s) + water \longrightarrow Na^+(aq) + OH^-(aq)$$

Animation
Dissolving ammonia in water

Ammonia gas consists of ammonia molecules. When ammonia gas dissolves in water, it reacts with water to give ammonium ions ($NH_4^+(aq)$) and hydroxide ions ($OH^-(aq)$). However, ammonia does not react with water completely. Only very few hydroxide ions are formed.

$$NH_3(g) + water \longrightarrow NH_3(aq)$$

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

Thus, aqueous ammonia contains ammonia molecules, water molecules and a few ammonium ions and hydroxide ions.

The properties of solutions of alkalis depend on the presence of mobile hydroxide ions ($OH^-(aq)$).

14.11 An introduction to analytical chemistry

The discipline of analytical chemistry consists of **qualitative** and **quantitative** analyses. The former deals with the identification of chemical species present in a sample; the latter deals with the determination of how much a constituent is present.

We can use the chemical reactions we have learnt in this unit to identify ions present in a sample (see Examples 14.1 and 14.2). N16

N16 Refer to page T19.

qualitative 定性　　quantitative 定量

Example 14.1

Q For each of the following pairs of species, suggest a chemical test to distinguish between them and write equation(s) of the reaction(s) involved.

a) $ZnCl_2(aq)$ and $CaCl_2(aq)$

b) $Ag_2O(s)$ and $CuO(s)$ — CE 2011 Paper 2 Q27

A a) Add dilute sodium hydroxide solution to each solution.

$CaCl_2(aq)$ gives a white precipitate which does not dissolve in excess alkali.

$Ca^{2+}(aq) + 2OH^-(aq) \longrightarrow Ca(OH)_2(s)$

$ZnCl_2(aq)$ gives a white precipitate which dissolves in excess alkali.

$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$

$Zn(OH)_2(s) + 2OH^-(aq) \longrightarrow [Zn(OH)_4]^{2-}(aq)$

b) Any one of the following:

- Heat each species strongly.

  Only $Ag_2O(s)$ decomposes to give a grey metal and a gas that relights a glowing splint.

  $2Ag_2O(s) \longrightarrow 4Ag(s) + O_2(g)$

- Heat each species with carbon strongly.

  Black $Ag_2O(s)$ changes to a grey metal.

  $2Ag_2O(s) + C(s) \longrightarrow 4Ag(s) + CO_2(g)$

  Black $CuO(s)$ changes to a reddish brown metal.

  $2CuO(s) + C(s) \longrightarrow 2Cu(s) + CO_2(g)$

Example 14.2

Q Consider the three substances listed below:

sodium carbonate solution, potassium chloride solution, dilute nitric acid

Suggest how you can distinguish the three substances and write ionic equations of the reactions involved.

A Add dilute sulphuric acid / hydrochloric acid to each substance.

Only the sodium carbonate solution gives effervescence.

$CO_3^{2-}(aq) + 2H^+(aq) \longrightarrow H_2O(l) + CO_2(g)$

Add silver nitrate solution to the remaining two substances.

Only the potassium chloride solution gives a white precipitate.

$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$

Practice 14.3

For each of the following pairs of substances, suggest a chemical test to distinguish one substance from the other and state the expected observations.

a) Calcium carbonate and calcium oxide

b) Ammonium chloride and potassium chloride

a) Add dilute hydrochloric acid to each substance separately.
Only calcium carbonate gives a gas that turns limewater milky (the gas is carbon dioxide).
b) Add dilute sodium hydroxide solution to each substance separately and warm.
Only ammonium chloride gives a gas that turns moist red litmus paper blue (the gas is ammonia).

Problem Solving

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Four labels have come off four reagent bottles.

Lead(II) nitrate $Pb(NO_3)_2$	Zinc chloride $ZnCl_2$
Magnesium nitrate $Mg(NO_3)_2$	Sodium chloride NaCl

Describe the tests you would do to identify which bottle contained which substance. State also your expected observations.

Problem Solving
Carry out a flame test on all the substances.
Only sodium chloride gives a golden yellow flame.
Dissolve the remaining three substances separately in water. Divide each solution into two portions.
Add silver nitrate solution to one portion of each solution.
Only the zinc chloride solution gives a white precipitate.
Add dilute sodium hydroxide solution to the remaining two solutions.
The lead(II) nitrate solution gives a white precipitate which is soluble in excess alkali.
The magnesium nitrate solution gives a white precipitate which does not dissolve in excess alkali.
(Other possible tests are also acceptable.)

14.12 Concentrated acids

Concentrated hydrochloric acid

Ordinary concentrated hydrochloric acid contains about 35% hydrogen chloride by mass. Its concentration is about 11 mol dm^{-3}.

This means 1 000 g of solution contains about 350 g of dissolved hydrogen chloride.

'mol dm^{-3}' is the unit of molarity. We will discuss this in Unit 15.

The reactions of concentrated hydrochloric acid are similar to those of dilute hydrochloric acid. However, the reaction is faster for the concentrated acid.

Concentrated nitric acid

Ordinary concentrated nitric acid contains about 70% nitric acid by mass. Its concentration is about 16 mol dm^{-3}. It tends to decompose to brown nitrogen dioxide gas. The decomposition of concentrated nitric acid is speeded up by light. So the acid is often kept in brown bottles (Fig. 14.26).

ASL 2008 Paper 1 Q2



Fig. 14.26 Concentrated nitric acid stored in a brown bottle

Reference Website
Refer to the following website for the use, manufacturing, storage and safety information of sulphuric acid: http://www.sulphuric-acid.org/ (accessed July 2014)

Concentrated nitric acid has the characteristics of a typical acid. For example, it reacts with a base to give a salt and water. However, it can also behave as an oxidizing agent and react with unreactive metals, such as copper.

We will discuss the oxidizing property of concentrated nitric acid in Topic 5 Redox Reactions, Chemical Cells and Electrolysis.

Concentrated sulphuric acid

Ordinary concentrated sulphuric acid contains about 98% sulphuric acid by mass. Its concentration is about 18 mol dm^{-3}.

Concentrated sulphuric acid has the characteristics of a typical acid. For example, it reacts with a base to give a salt and water. However, it can also behave as an oxidizing agent and react with unreactive metals, such as copper.

We will discuss the oxidizing property of concentrated sulphuric acid in Topic 5 Redox Reactions, Chemical Cells and Electrolysis.

14.13 Corrosive nature of concentrated acids and alkalis

Caution: Never let concentrated acids and alkalis get on your skin. You must always wear safety glasses and protective gloves when working with them.

Concentrated acids are very corrosive. They can 'eat away' metals, skin and cloth (Fig. 14.27).

CE 2009 Paper 1 Q4(a)
CE 2010 Paper 2 Q42
DSE 2014 Paper 1A Q15



Fig. 14.27 Concentrated acids corrode other substances upon contact

DSE 2013 Paper 1A Q9

Concentrated alkalis are also very corrosive. Sodium hydroxide is often called caustic soda (caustic means burning). Concentrated alkalis do more damage to the skin and eyes than concentrated acids do.

Video
Action of concentrated sodium hydroxide solution on meat

Worksheet Assessment Form
A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Find & Share

Corrosive nature of concentrated acids and alkalis

Tasks

1 Choose ONE of the following chemicals and search for the information about its hazardous nature.

- concentrated hydrochloric acid
- concentrated nitric acid
- concentrated sulphuric acid
- concentrated sodium hydroxide solution
- concentrated potassium hydroxide solution

2 Design a poster for the chemical in task 1 to alert people about its hazardous nature. You should include the following information:

- hazard warning label;
- risk description; and
- safety precautions.

Reference websites

1 You may start your search from the following website:

- The Chemical Database from the University of Akron

  http://ull.chemistry.uakron.edu/erd/

2 Extend your search using search engines and appropriate keywords.

Remarks

Some websites use different spelling when naming chemicals, for example, sulphuric acid / sulfuric acid.

Refer to page T20.

14.14 Hygroscopic and deliquescent substances

When we allow a beaker of concentrated sulphuric acid to stand in the air, the volume of the liquid in the beaker increases. This is because concentrated sulphuric acid is **hygroscopic** (Fig. 14.28). Hygroscopic substances are substances that can absorb water vapour from the air.

CE 2008 Paper 2 Q38

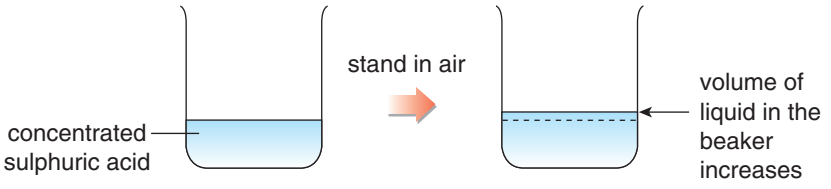


Fig. 14.28 Concentrated sulphuric acid is a hygroscopic substance

Other hygroscopic substances include calcium oxide, **anhydrous** copper(II) sulphate and silica gel.

When we allow sodium hydroxide to stand in the air, it absorbs water vapour from the air to such an extent that it dissolves in the absorbed water and forms a solution. Sodium hydroxide is a **deliquescent** substance (Fig. 14.29). Other deliquescent substances are potassium hydroxide and anhydrous calcium chloride.



Fig. 14.29 Sodium hydroxide is a deliquescent substance

hygroscopic 吸濕的 anhydrous 無水的 deliquescent 潮解的

N17 Refer to page T22.

Drying agents N17

Hygroscopic and deliquescent substances (e.g. concentrated sulphuric acid and anhydrous calcium chloride) can absorb water vapour from the air. Therefore they are often used as drying agents (Fig. 14.30).

Notice the end of the delivery tube for the incoming gas is put under the concentrated sulphuric acid. This is to ensure that all the incoming gas can pass through the drying agent.

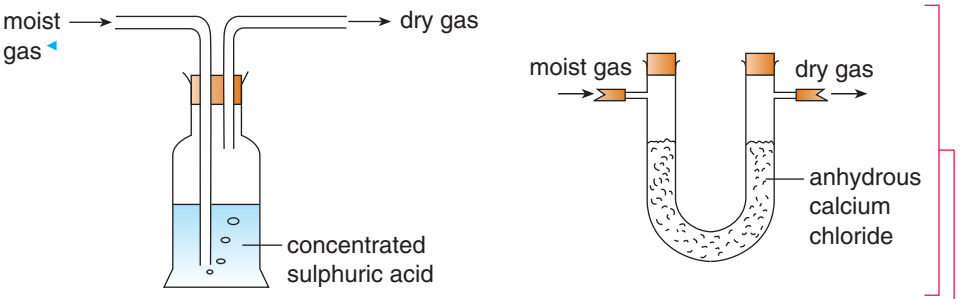


Fig. 14.30 Concentrated sulphuric acid and anhydrous calcium chloride can be used to dry gases

CE 2010 Paper 2 Q5

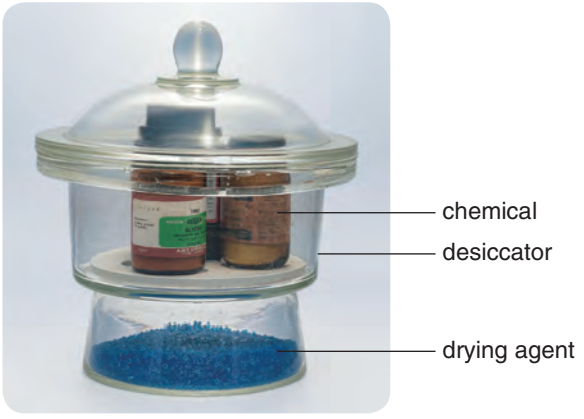


Fig. 14.31 Using drying agents to dry chemicals in a *desiccator*

N18 Calcium oxide cannot be used to dry hydrogen chloride gas because calcium oxide reacts with hydrogen chloride gas. Concentrated sulphuric acid or anhydrous calcium chloride can be used to dry hydrogen chloride gas.

Notice that a drying agent should not react with the material being dried. For example, we do not use anhydrous calcium chloride and concentrated sulphuric acid to dry ammonia gas because ammonia reacts with both of them. We can use calcium oxide to dry ammonia. N18

CE 2008 Paper 2 Q38
CE 2011 Paper 2 Q20

desiccator 乾燥器

Have you mastered?

Key terms

citric acid 檸檬酸 2 carbonic acid 碳酸 2 phosphoric acid 磷酸 2 acidic 酸性的 2

ethanoic acid 乙酸 2 hydrochloric acid 氫氯酸 2 sulphuric acid 硫酸 3 nitric acid 硝酸 3

lactic acid 乳酸 3 indicator 指示劑 5 litmus solution 石蕊試液 5 litmus paper 石蕊試紙 5

methyl orange 甲基橙試液 5 phenolphthalein 酚酞試液 5 effervescence 泡騰 7 base 鹽基 9

neutralization 中和作用 9 dissociation 離解作用 11 hydroxonium ion 水合氫離子 12

basicity 鹽基度 13 monobasic acid 一元酸 13 dibasic acid 二元酸 13

tribasic acid 三元酸 14 alkali 鹼 14 alkaline 鹼性的 14 acidity 酸度 15

potassium hydroxide 氫氧化鉀 15 sodium hydroxide 氫氧化鈉 15

calcium hydroxide 氫氧化鈣 15 aqueous ammonia 氨水 15 complex salt 絡鹽 18

qualitative analysis 定性分析 21 quantitative analysis 定量分析 21 hygroscopic 吸濕的 26

anhydrous 無水的 26 deliquescent 潮解的 26 drying agent 乾燥劑 27

Checklist

After studying this unit, you should be able to

- ☐ describe the characteristics of acids and their typical reactions;
- ☐ write chemical and ionic equations for the reactions of acids;
- ☐ describe the role of water for acids to exhibit their properties;
- ☐ describe the characteristics of alkalis and their typical reactions;
- ☐ write chemical and ionic equations for the reactions of alkalis;
- ☐ describe the corrosive nature of acids and alkalis;
- ☐ select an appropriate test to detect the presence of a chemical species;
- ☐ select an appropriate test to distinguish between different species.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 An acid is a hydrogen-containing substance that gives hydrogen ions ($H^+(aq)$) as the only type of positive ions when dissolved in water.

2 The following table summarizes the characteristics of dilute acids.

Characteristics of dilute acids	Example
a) Most dilute acids have a sour taste	—
b) Effect on indicators	**Indicator** — **Colour in an acid** Litmus solution — red Methyl orange — red Phenolphthalein — colourless
c) Reaction with metals metal + dilute acid ⟶ salt + hydrogen	$Mg(s) + 2HCl(aq) \longrightarrow MgCl_2(aq) + H_2(g)$ or $Mg(s) + 2H^+(aq) \longrightarrow Mg^{2+}(aq) + H_2(g)$
d) Reaction with carbonate carbonate + dilute acid ⟶ salt + water + carbon dioxide	$Na_2CO_3(s \text{ or } aq) + 2HCl(aq) \longrightarrow 2NaCl(aq) + H_2O(l) + CO_2(g)$ or $CO_3^{2-}(aq) + 2H^+(aq) \longrightarrow H_2O(l) + CO_2(g)$
e) Reaction with hydrogencarbonate hydrogencarbonate + dilute acid ⟶ salt + water + carbon dioxide	$NaHCO_3(s \text{ or } aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$ or $HCO_3^-(aq) + H^+(aq) \longrightarrow H_2O(l) + CO_2(g)$
f) Reaction with hydroxides and oxides of metals acid + metal hydroxide ⟶ salt + water acid + metal oxide ⟶ salt + water	$2HCl(aq) + Mg(OH)_2(s) \longrightarrow MgCl_2(aq) + 2H_2O(l)$ $H_2SO_4(aq) + CuO(s) \longrightarrow CuSO_4(aq) + H_2O(l)$
g) All dilute acids conduct electricity due to the presence of mobile ions	—

3 Water must be present for an acid to show its acidic properties.

4 The maximum number of hydrogen ions produced by an acid molecule is called the basicity of the acid.

5 A base is a compound which reacts with an acid to give a salt and water only. Bases are usually oxides or hydroxides of metals.

acid + base ⟶ salt + water

Have you mastered?

6 Alkalis are bases that are soluble in water.

7 The properties of solutions of alkalis depend on the presence of mobile hydroxide ions ($OH^-(aq)$).

8 Characteristics of solutions of alkalis:

a) Solutions of alkalis have a bitter taste.

b) Dilute solutions of alkalis have a 'slippery' feel.

c) Colours of indicators in the solution of an alkali:

Indicator	Colour in the solution of an alkali
Litmus solution	blue
Methyl orange	yellow
Phenolphthalein	red

d) Adding dilute sodium hydroxide solution to solutions containing some metal ions gives precipitates as shown in the following table:

Adding NaOH(aq) to solution containing	Colour of precipitate formed	Precipitate dissolves in excess NaOH(aq)?	Ionic equation(s)
$Ca^{2+}(aq)$	white	✘	$Ca^{2+}(aq) + 2OH^-(aq) \longrightarrow Ca(OH)_2(s)$
$Mg^{2+}(aq)$	white	✘	$Mg^{2+}(aq) + 2OH^-(aq) \longrightarrow Mg(OH)_2(s)$
$Al^{3+}(aq)$	white	✔ (a colourless solution forms)	$Al^{3+}(aq) + 3OH^-(aq) \longrightarrow Al(OH)_3(s)$ $Al(OH)_3(s) + OH^-(aq) \longrightarrow [Al(OH)_4]^-(aq)$
$Pb^{2+}(aq)$	white	✔ (a colourless solution forms)	$Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$ $Pb(OH)_2(s) + 2OH^-(aq) \longrightarrow [Pb(OH)_4]^{2-}(aq)$
$Zn^{2+}(aq)$	white	✔ (a colourless solution forms)	$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$ $Zn(OH)_2(s) + 2OH^-(aq) \longrightarrow [Zn(OH)_4]^{2-}(aq)$
$Fe^{2+}(aq)$	green	✘	$Fe^{2+}(aq) + 2OH^-(aq) \longrightarrow Fe(OH)_2(s)$
$Fe^{3+}(aq)$	reddish brown	✘	$Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$
$Cu^{2+}(aq)$	pale blue	✘	$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$

Have you mastered?

e) Adding dilute aqueous ammonia to solutions containing some metal ions gives precipitates as shown in the following table:

Adding $NH_3(aq)$ to solution containing	Colour of precipitate formed	Precipitate dissolves in excess $NH_3(aq)$?	Ionic equation(s)
$Mg^{2+}(aq)$	white	✘	$Mg^{2+}(aq) + 2OH^-(aq) \longrightarrow Mg(OH)_2(s)$
$Al^{3+}(aq)$	white	✘	$Al^{3+}(aq) + 3OH^-(aq) \longrightarrow Al(OH)_3(s)$
$Pb^{2+}(aq)$	white	✘	$Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$
$Zn^{2+}(aq)$	white	✔ (a colourless solution forms)	$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$ $Zn(OH)_2(s) + 4NH_3(aq) \longrightarrow [Zn(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$
$Fe^{2+}(aq)$	green	✘	$Fe^{2+}(aq) + 2OH^-(aq) \longrightarrow Fe(OH)_2(s)$
$Fe^{3+}(aq)$	reddish brown	✘	$Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$
$Cu^{2+}(aq)$	pale blue	✔ (a deep blue solution forms)	$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$ $Cu(OH)_2(s) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$

f) Heating solids or solutions of ammonium compounds with solutions of alkalis liberates ammonia gas.

e.g. $NH_4Cl(s \text{ or } aq) + NaOH(aq) \xrightarrow{\text{heat}} NH_3(g) + NaCl(aq) + H_2O(l)$

or

$NH_4^+(aq) + OH^-(aq) \xrightarrow{\text{heat}} NH_3(g) + H_2O(l)$

g) A solution of alkali reacts with an acid to give a salt and water only.

e.g. $NaOH(aq) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l)$

or

$OH^-(aq) + H^+(aq) \longrightarrow H_2O(l)$

h) All solutions of alkalis conduct electricity due to the presence of mobile ions.

9 Concentrated acids and alkalis are corrosive.

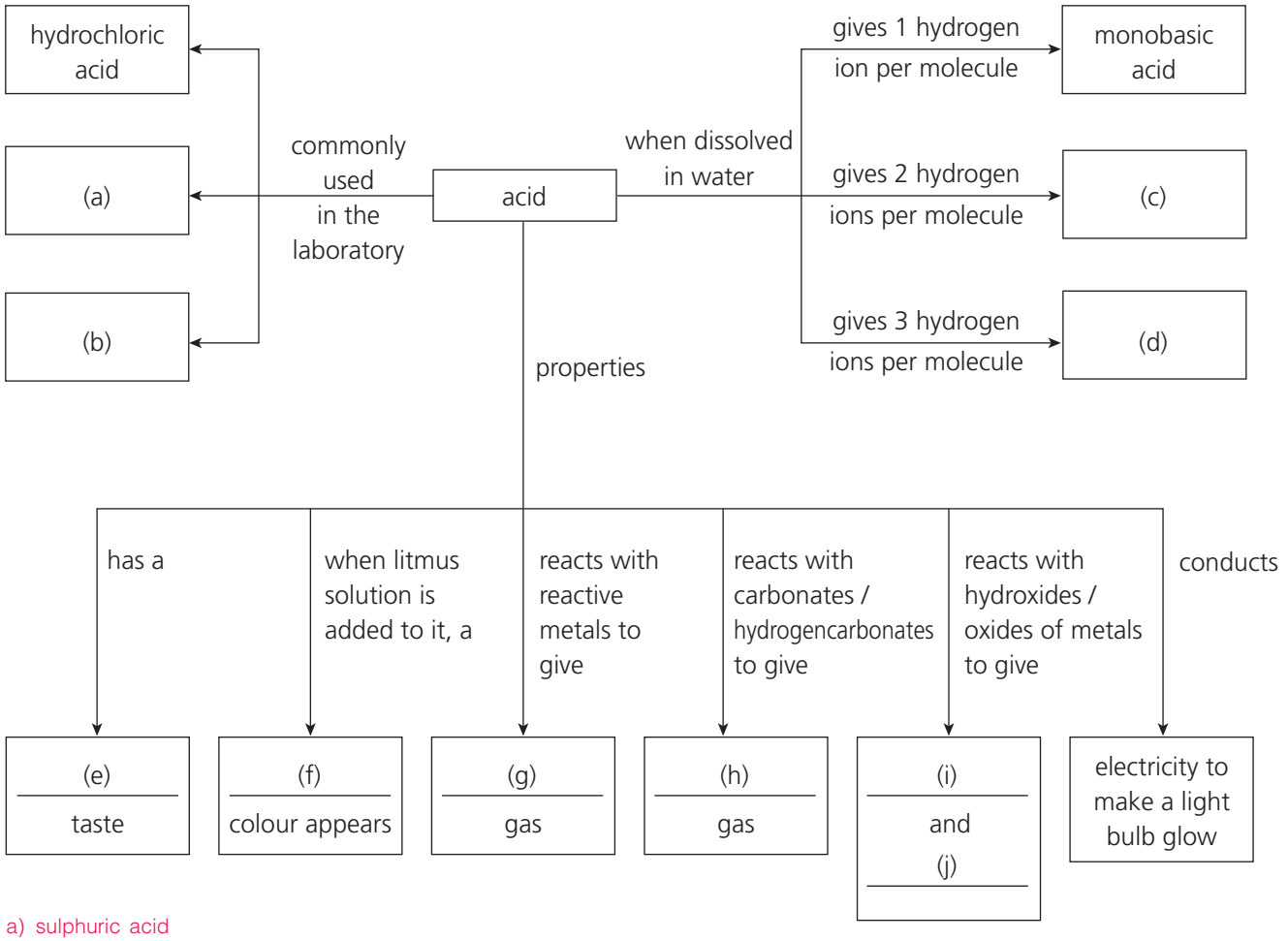
10 Anhydrous calcium chloride and concentrated sulphuric acid are common drying agents.

Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

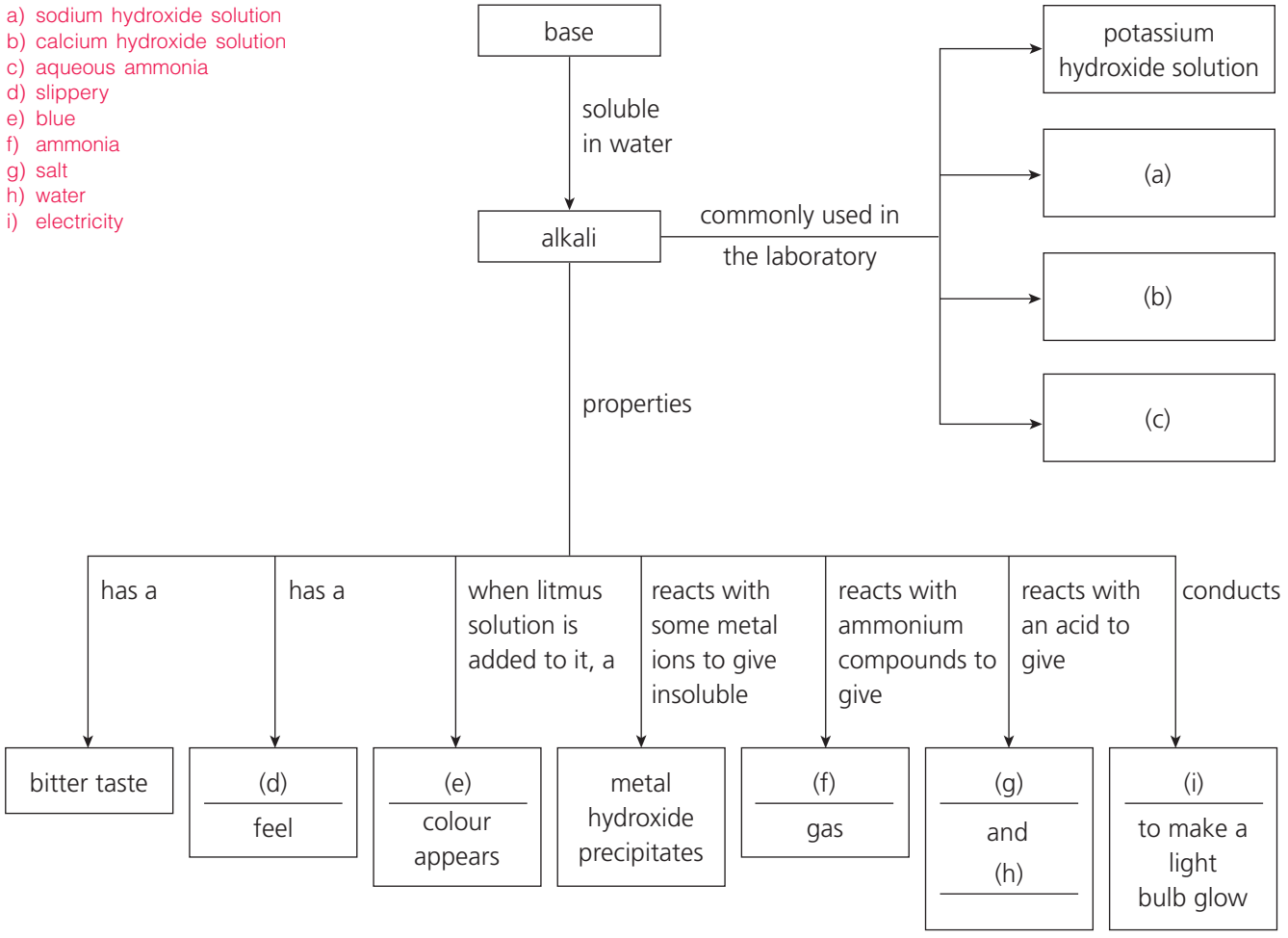
1 Complete the following concept map.



a) sulphuric acid
b) nitric acid
c) dibasic acid
d) tribasic acid
e) sour
f) red
g) hydrogen
h) carbon dioxide
i) salt
j) water

2 Complete the following concept map.

a) sodium hydroxide solution
b) calcium hydroxide solution
c) aqueous ammonia
d) slippery
e) blue
f) ammonia
g) salt
h) water
i) electricity



3 Complete the following table.

Indicator	Colour in	
	dilute hydrochloric acid	dilute sodium hydroxide solution
Litmus solution	red	blue
Methyl orange	red	yellow
Phenolphthalein	colourless	red

4 Complete the following table concerning the addition of dilute sodium hydroxide solution to solutions containing some metal ions.

Adding NaOH(aq) to solution containing	Colour of precipitate formed	Precipitate dissolves in excess NaOH(aq)? (Yes or No)	Ionic equation(s)
$Ca^{2+}(aq)$	white	No	$Ca^{2+}(aq) + 2OH^-(aq) \longrightarrow Ca(OH)_2(s)$
$Mg^{2+}(aq)$	white	No	$Mg^{2+}(aq) + 2OH^-(aq) \longrightarrow Mg(OH)_2(s)$
$Al^{3+}(aq)$	white	Yes	$Al^{3+}(aq) + 3OH^-(aq) \longrightarrow Al(OH)_3(s)$ $Al(OH)_3(s) + OH^-(aq) \longrightarrow [Al(OH)_4]^-(aq)$
$Pb^{2+}(aq)$	white	Yes	$Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$ $Pb(OH)_2(s) + 2OH^-(aq) \longrightarrow [Pb(OH)_4]^{2-}(aq)$
$Zn^{2+}(aq)$	white	Yes	$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$ $Zn(OH)_2(s) + 2OH^-(aq) \longrightarrow [Zn(OH)_4]^{2-}(aq)$
$Fe^{2+}(aq)$	green	No	$Fe^{2+}(aq) + 2OH^-(aq) \longrightarrow Fe(OH)_2(s)$
$Fe^{3+}(aq)$	reddish brown	No	$Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$
$Cu^{2+}(aq)$	pale blue	No	$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$

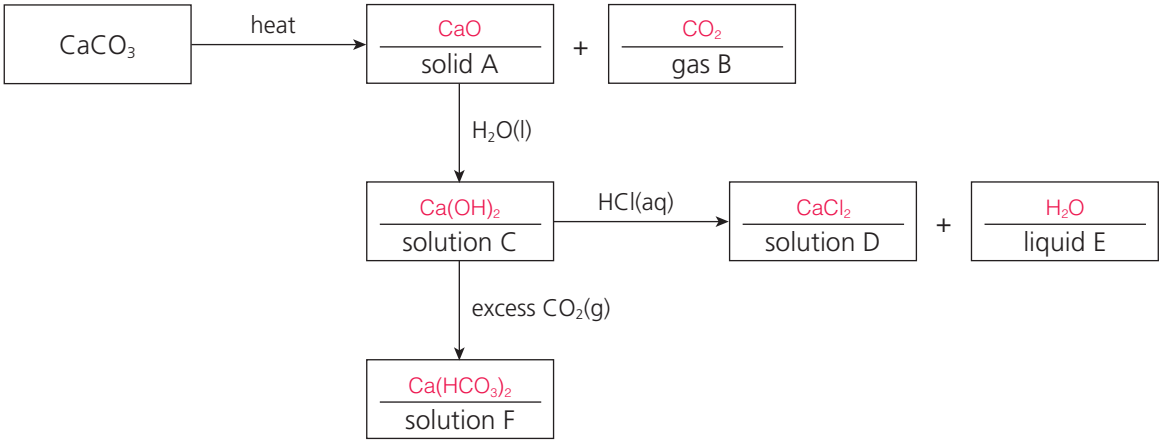
5 Complete the following table concerning the addition of dilute aqueous ammonia to solutions containing some metal ions.

Adding $NH_3(aq)$ to solution containing	Colour of precipitate formed	Precipitate dissolves in excess $NH_3(aq)$? (Yes or No)	Ionic equation(s)
$Mg^{2+}(aq)$	white	No	$Mg^{2+}(aq) + 2OH^-(aq) \longrightarrow Mg(OH)_2(s)$
$Al^{3+}(aq)$	white	No	$Al^{3+}(aq) + 3OH^-(aq) \longrightarrow Al(OH)_3(s)$
$Pb^{2+}(aq)$	white	No	$Pb^{2+}(aq) + 2OH^-(aq) \longrightarrow Pb(OH)_2(s)$
$Zn^{2+}(aq)$	white	Yes	$Zn^{2+}(aq) + 2OH^-(aq) \longrightarrow Zn(OH)_2(s)$ $Zn(OH)_2(s) + 4NH_3(aq) \longrightarrow [Zn(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$
$Fe^{2+}(aq)$	green	No	$Fe^{2+}(aq) + 2OH^-(aq) \longrightarrow Fe(OH)_2(s)$
$Fe^{3+}(aq)$	reddish brown	No	$Fe^{3+}(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s)$
$Cu^{2+}(aq)$	pale blue	Yes	$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$ $Cu(OH)_2(s) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$

6 This question looks at the chemistry of Group 2 elements and their compounds.

The flowchart below shows some reactions involving calcium compounds.

Identify substances A–F by writing their formulae in the boxes.



(OCR Advanced Subsidiary GCE, Chemistry, 2811/01, Jun. 2008, 4(a))

Part II Multiple choice questions

7 The table shows some possible uses of ethanoic acid, citric acid and phosphoric acid. Which row of the table is correct?

	Use of ethanoic acid	Use of citric acid	Use of phosphoric acid
A	as food flavouring	in rust remover	as vinegar
B	in rust remover	as food flavouring	as vinegar
C	in rust remover	as vinegar	as food flavouring
(D)	as vinegar	as food flavouring	in rust remover

(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Jun. 2010, 36)

8 $CuO(s)$ is insoluble in water and there is no separate Cu^{2+} ion or O^{2-} ion in water.

8 The equation representing the reaction between copper(II) oxide and dilute sulphuric acid is

$$CuO(s) + H_2SO_4(aq) \longrightarrow CuSO_4(aq) + H_2O(l)$$

The ionic equation for the reaction is

A $Cu^{2+}(s) + SO_4^{2-}(aq) \longrightarrow CuSO_4(aq)$
B $O^{2-}(s) + H_2SO_4(aq) \longrightarrow H_2O(l) + SO_4^{2-}(aq)$
(C) $CuO(s) + 2H^+(aq) \longrightarrow Cu^{2+}(aq) + H_2O(l)$
D $CuO(s) + H_2SO_4(aq) \longrightarrow Cu^{2+}SO_4^{2-}(aq) + H_2O(l)$

(Edexcel Advanced Subsidiary GCE, Unit 1, May 2012, 18)

9 Which of these would NOT react with dilute hydrochloric acid?

(A) Calcium chloride
B Calcium oxide
C Calcium hydroxide
D Calcium carbonate

(Edexcel GCSE (Higher Tier), Chemistry, Unit C1a, Jun. 2012, 38)

10 Which of the following compounds has the *highest* basicity?

A HNO_3
B H_2CO_3
C H_3PO_4 H_3PO_4 is a tribasic acid.
D CH_3COOH

11 Which of the following statements concerning concentrated sulphuric acid is INCORRECT?

A Concentrated sulphuric acid can be used as a drying agent for ammonia.
B Adding concentrated sulphuric acid to sugar will give a steamy fume.
C Blue litmus paper will finally turn black when dropped into concentrated sulphuric acid.
D When a beaker of concentrated sulphuric acid is left in air, the volume of liquid inside the beaker increases gradually.

(HKCEE, Paper 2, 2008, 38)

12 Compound X is soluble in water. Addition of dilute aqueous ammonia to a solution of X gives a precipitate. The precipitate does not dissolve upon the addition of excess alkali. X may be

A $CuCl_2$. $FeSO_4$ is soluble in water. Addition of dilute aqueous ammonia to $FeSO_4(aq)$ gives a precipitate, iron(II) hydroxide.
B $FeSO_4$.
C $PbSO_4$. $Fe^{2+}(aq) + 2OH^-(aq) \longrightarrow Fe(OH)_2(s)$
D $ZnCl_2$. The precipitate does not dissolve in excess alkali.

13 A mixture of $(NH_4)_2SO_4(aq)$ and $MgSO_4(aq)$ is heated with excess NaOH(aq). Which of the following observations is correct?

A No pungent gas is evolved and no precipitate is formed.
B No pungent gas is evolved but a white precipitate is formed.
C A pungent gas is evolved but no precipitate is formed.
D A pungent gas is evolved and a white precipitate is formed.

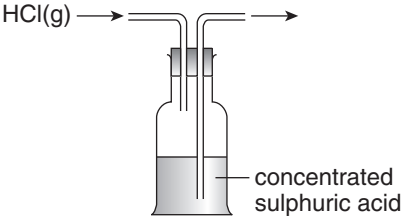
(HKCEE, Paper 2, 2010, 20)

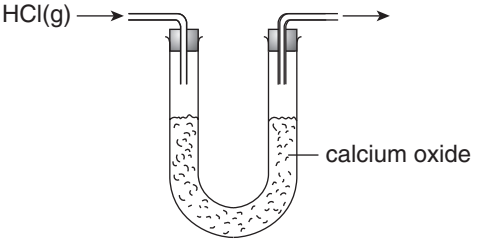
14 Which of the following reagents would enable you to separate iron(III) hydroxide from a mixture of iron(III) hydroxide and copper(II) hydroxide?

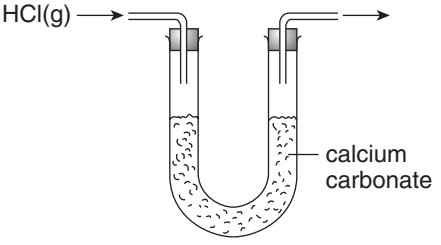
A Dilute hydrochloric acid
B Aqueous ammonia
C Dilute nitric acid
D Sodium hydroxide solution

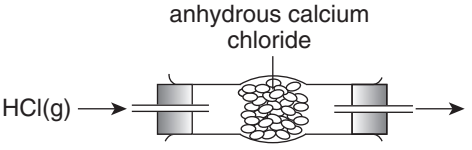
(Edexcel Advanced GCE, Unit 5, Feb. 2012, 6)

15 Which of the following set-ups can be used to dry hydrogen chloride gas?

A


B


C


D


(HKCEE, Paper 2, 2010, 5)

14 Iron(III) hydroxide is insoluble in aqueous ammonia.
Copper(II) hydroxide reacts with excess aqueous ammonia to form a soluble complex salt.
$Cu(OH)_2(s) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{2+}(aq) + 2OH^-(aq)$

16 In which of the following processes would a colourless gas evolve?

(1) Magnesium is added to dilute sulphuric acid.
(2) Ammonium chloride is heated with calcium hydroxide.
(3) Water is added to a solid mixture of citric acid and sodium hydrogencarbonate.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKDSE, Paper 1A, 2012, 19)

17 Which of the following methods can be used to distinguish between $ZnCl_2(aq)$ and $CaBr_2(aq)$?

(1) Adding $NH_3(aq)$
(2) Performing a flame test
(3) Evaporating to dryness

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKDSE, Paper 1A, 2012, 20)

Part III Structured questions

Answers for the questions in Hong Kong public examinations are not provided.

18 A student proposed the following methods to accomplish three tasks. The proposed methods were all considered inappropriate.

Task	Proposed method
1 To prepare hydrogen gas from a metal and an acid.	Add copper to dilute hydrochloric acid.
2 To prepare carbon dioxide gas from an acid.	Add calcium carbonate to dilute sulphuric acid.
3 To choose an active ingredient for an antacid.	Use sodium hydroxide as the active ingredient.

For each task,

a) explain why the proposed method is inappropriate; and

b) suggest a proper method.

19 For each of the following experiments,

i) state an expected observation; and

ii) write a chemical or ionic equation for the reaction involved.

a) Adding dilute sulphuric acid to zinc granules

b) Adding dilute nitric acid to solid copper(II) carbonate

c) Adding dilute hydrochloric acid to solid sodium hydrogencarbonate

20 Explain why water should NOT be added to concentrated H_2SO_4 in order to dilute the acid.

(HKASLE, Paper 1, 2009, 4(a))

Unit Exercise

18 (1) a) Copper does not react with dilute hydrochloric acid.
b) Add magnesium to dilute hydrochloric acid.
(2) a) Insoluble calcium sulphate is formed when calcium carbonate reacts with dilute sulphuric acid. The calcium sulphate covers the surface of calcium carbonate and prevents further reaction.
b) Add calcium carbonate to dilute hydrochloric acid / dilute nitric acid.
(3) a) Sodium hydroxide is very corrosive.
b) Use magnesium hydroxide / aluminium hydroxide as the active ingredient.

19 a) Gas bubbles are given off. / The zinc granules dissolve in the acid.
$Zn(s) + H_2SO_4(aq) \longrightarrow ZnSO_4(aq) + H_2(g)$
or $Zn(s) + 2H^+(aq) \longrightarrow Zn^{2+}(aq) + H_2(g)$
b) Effervescence occurs. / Solid copper(II) carbonate dissolves in the acid. / A blue solution is formed.
$CuCO_3(s) + 2HNO_3(aq) \longrightarrow Cu(NO_3)_2(aq) + H_2O(l) + CO_2(g)$
or $CuCO_3(s) + 2H^+(aq) \longrightarrow Cu^{2+}(aq) + H_2O(l) + CO_2(g)$
c) Effervescence occurs. / Solid sodium hydrogencarbonate dissolves in the acid.
$NaHCO_3(s) + HCl(aq) \longrightarrow NaCl(aq) + H_2O(l) + CO_2(g)$
or $NaHCO_3(s) + H^+(aq) \longrightarrow Na^+(aq) + H_2O(l) + CO_2(g)$

21 This question is concerned with the reactions of acids.

Refer to page T22.

a) Define an acid.

b) The table below gives data about some solid compounds of copper and sodium. Use the data to help to answer the questions in (i) and (ii).

Compound	Formula	Colour	Solubility in water	Colour in solution
Copper(II) oxide	CuO	black	insoluble	—
Copper(II) chloride	$CuCl_2$	yellow	soluble	blue / green
Copper(II) carbonate	$CuCO_3$	green	insoluble	—
Sodium chloride	NaCl	white	soluble	colourless
Sodium carbonate	Na_2CO_3	white	soluble	colourless

i) Write equations, including state symbols, for the following reactions of hydrochloric acid.

copper(II) oxide and hydrochloric acid

sodium carbonate and hydrochloric acid

ii) What would you expect to see when copper(II) carbonate reacts with hydrochloric acid?

(OCR Advanced Subsidiary GCE, Chemistry, 2813/01, Jan. 2008, 3(a)–(b))

22 For each of the following experiments,

Refer to page T23.

i) state the expected observations; and

ii) write chemical or ionic equation(s) for the reaction(s) involved.

a) Adding dilute aqueous ammonia to copper(II) sulphate solution

b) Adding dilute aqueous ammonia to aluminium nitrate solution

c) Adding dilute sodium hydroxide solution to zinc chloride solution

d) Warming ammonium chloride solution with calcium hydroxide solution

23 X is a white solid that dissolves readily in water to form a colourless solution. X contains one cation and one anion. Complete the table below:

Test	Observation	Inference
1 Warm solid X with __(a)__ dilute sodium hydroxide solution / calcium hydroxide	A gas that can turn moist red litmus paper blue is evolved.	The gas is __(b)__. X probably contains __(c)__ ion.
2 Add dilute nitric acid followed by silver nitrate solution to a solution of X.	A white precipitate is formed. b) ammonia c) ammonium d) silver chloride e) chloride	The white precipitate is __(d)__. X probably contains __(e)__ ion.

24 White solid Y contains one cation and one anion. A student conducts some tests on Y or its solution. Complete the table below:

Test	Observation	Inference
1 Carry out a flame test on Y.	__(a)__	Y probably contains calcium ion.
2 i) Add dilute hydrochloric acid to Y.	__(b)__ occurs. A colourless solution is formed.	A gas is evolved.
ii) Pass the gas through limewater.	The limewater turns milky.	The gas is __(c)__. Y probably contains __(d)__ ion.
3 Add dilute sodium hydroxide solution to a solution of Y.	__(e)__	The new substance observed is __(f)__.

24 a) A brick-red flame is observed.
b) Effervescence
c) carbon dioxide
d) carbonate
e) A white precipitate is formed.
f) calcium hydroxide

25 The information in the box was on the Internet.
Refer to page T23.

Breathtaking Finish
Tim Henman used smelling salts to help revive his fortunes at Wimbledon yesterday.



The active chemical in smelling salts is ammonium carbonate, $(NH_4)_2CO_3$.

Describe how smelling salts can be tested to show that they contain ammonium ions and carbonate ions. Give the results of the tests.

a) Test and result for ammonium ions.

b) Test and result for carbonate ions.

(AQA GCSE (Higher Tier), Chemistry, Unit C3, Jun. 2008, 3(a)–(b))

26 For each of the following pairs of substances, suggest a chemical test to distinguish one substance from the other and state the expected observations.
Refer to page T23.

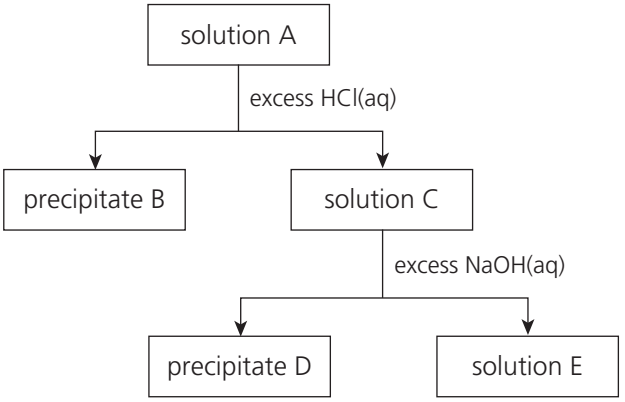
a) Aluminium nitrate solution and calcium nitrate solution

b) Iron(II) sulphate solution and iron(III) sulphate solution

27 An aqueous solution A is known to contain the following three cations:

$NH_4^+(aq)$, $Cu^{2+}(aq)$ and $Ag^+(aq)$

The flow diagram below outlines a series of tests that can be used to detect the presence of two of the above cations in A.



a) Write an ionic equation for the formation of B from A.
$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$

b) Name a process for separating C from B.
Filtration

c) i) Name D.
Copper(II) hydroxide

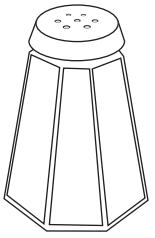
ii) What is the colour of D?
Pale blue

d) E contains one of the above cations. Suggest how you can show experimentally the presence of the cation.
Warm E. A gas that turns moist red litmus paper blue is evolved. The gas is ammonia. This shows the presence of ammonium ions in E.

e) Suggest a colour for A. Explain your answer.
Blue; the solution contains blue $Cu^{2+}(aq)$ ions and the other ions are colourless.

28 The label is from a packet of Low Sodium Salt.

LOW SODIUM SALT



INGREDIENTS
potassium chloride
sodium chloride
Anti-caking agent: magnesium carbonate

A student tested some Low Sodium Salt to show that it contains carbonate ions and chloride ions.

a) Describe and give the result of a test for carbonate ions.
Add dilute hydrochloric acid to the salt.
A gas that can turn limewater milky is evolved.

b) A student identified chloride ions using acidified silver nitrate solution.

State what you would see when acidified silver nitrate solution is added to a solution of Low Sodium Salt.
A white precipitate

c) Flame tests can be used to identify potassium ions and sodium ions.

Suggest why it is difficult to identify BOTH of these ions in Low Sodium Salt using a flame test.

(AQA GCSE (Higher Tier), Chemistry, Unit C3, Jan. 2009, 1(a))

28 c) The presence of potassium cannot be shown. As in a flame test, the lilac flame of potassium will be masked by the brilliant golden yellow flame of sodium.